Neuron mechanisms III: where the degree-16 parity lives

`wd3_k0_N2048`, step 500k — post-nonlinearity units, uniform evaluation data

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Setup

We use the same protocol as in the $d4/d8$ reports: for each neuron, $a = \phi(z)$ with $\phi = \text{relu}(\tanh)$ and $z = (Vh)_n$, evaluated on all 2^{16} uniform inputs. A neuron is called $d16$-*dominant* when χ_{d16} is the largest among its 15 tracked tree-mode Walsh coefficients. To understand what drives each neuron, we decompose $\text{Var}(z)$ into contributions from the 16 raw singleton inputs, the tree degree-2/4 modes, the two half-parities $y_0 = \chi_{d8_0}$, $y_1 = \chi_{d8_1}$, and χ_{d16} itself.

Headline: the target parity has no dedicated neurons

Only **25 of 8192** neurons are $d16$-dominant (6/10/3/6 per block), but these neurons do not carry an appreciable fraction of the actual $d16$ computation.

(i) 19 wide raw-bit gates (blocks 0–2). Their pre-activation z is built almost entirely from *all sixteen* singleton inputs ($\geq 99\%$ of $\text{Var}\, z$). They are the width-16 version of the same balanced-input majority-gate mechanism seen at lower degrees. At width 16, however, the neuron's Walsh spectrum is distributed over so many modes that each individual tree coefficient is extremely small. Their tree coverage is only 0.007, continuing the sequence $d2 : 1/3 \to d4 : 0.15 \to d8 : 0.044 \to d16 : 0.007$, and their $d16$ power is only $a^2(d16) \sim 10^{-5}$–10^{-4}. Thus their apparent "$d16$-dominance" mostly comes from taking an argmax over fifteen coefficients that are all close to zero.

(ii) 6 near-dead receivers (block 3). These neurons do have a genuine $d16$-shaped component: about 40% of their activation variance is in χ_{d16}, and their tree coverage is ≈ 0.95. But their total activation variance is only $\text{Var}(a) \sim 10^{-6}$, so their contribution to the network-level computation is negligible.

Who actually carries χ_{d16}

The layer-level accounting over *all* neurons (Fig. 1, middle) makes the answer clear. The total degree-16 power $H(d16)$ is 1.6×10^{-3} and 1.4×10^{-3} in blocks 0–1, which is essentially the noise floor. It then rises to 2.8×10^{-2} in block 2 and 1.3×10^{-1} in block 3.

At exactly these blocks, almost all of the $d16$ signal is carried by the 8+8 **corner units identified in the** $d8$ **study**: they account for 98.7% of $H(d16)$ in block 2 and 99.8% in block 3. There are 189 such neurons. Their input has the form $\gamma(\pm y_0 \pm y_1)$, where y_0 and y_1 are the two degree-8 half-parities, so after the nonlinearity they act as two-input corner detectors. As in the degree-2 case, each such corner detector has equal nonconstant Walsh power in its two inputs and in their product. Here that means equal thirds over $\{\chi_{d8_0}, \chi_{d8_1}, \chi_{d16}\}$.

By contrast, the 25 neurons labeled "$d16$-dominant" contribute only 0.2% of $H(d16)$ in block 2 and 0.04% in block 3.

Unlike $d8$ — which is first created in blocks 0–1 and then passed to dedicated *amplifier* neurons — no analogous $d16$ amplifier population appears. The reason is architectural: $d16$ is created only in blocks 2–3, immediately before the final linear readout uses the residual stream. There is therefore no later residual block that needs to extract and amplify a cleaner copy of the feature.

As a result, the degree-16 parity remains a *collective* representation all the way to the readout. It appears as one component of many 8+8 corner neurons, rather than becoming the main computation of a dedicated neuron.

Takeaway. The mechanism across the full parity hierarchy is now consistent. Degree-2, 4, and 8 modes are generated by the same basic template: balanced linear combinations passed through relu(tanh). When the inputs

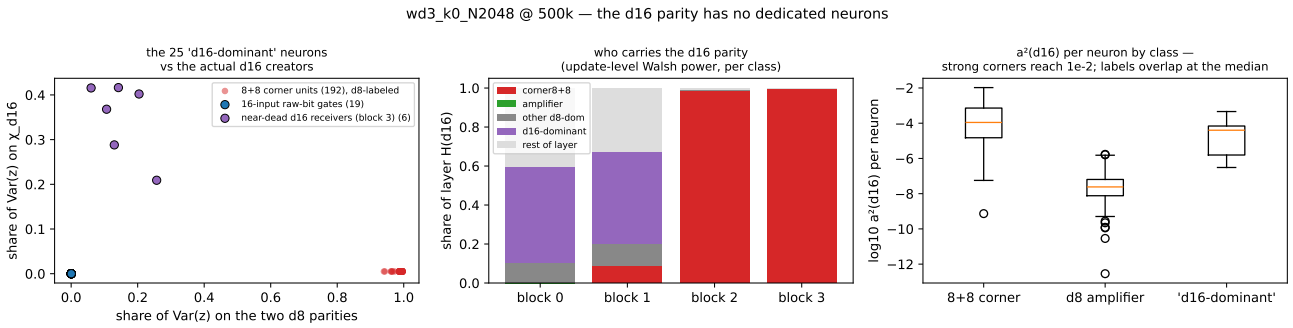


Figure 1: Left: the 25 $d16$-dominant neurons in input space — 19 wide raw-bit gates at the origin, 6 near-dead receivers; the actual creators (red, $d8$-labeled corner units) sit at $x \approx 1$. Middle: share of layer $H(d16)$ by class — at the blocks where $H(d16)$ is macroscopic (2.8×10^{-2}, 1.3×10^{-1}), the corner units own it (98.7%, 99.8%). Right: per-neuron $a^2(d16)$ by class — the strongest corners reach 10^{-2}, two orders above anything "$d16$-dominant".

class	input (z)	blocks	n	tree cov.	share of $H(d16)$ @ b2/b3
8+8 corner ($d8$-labeled)	$\gamma(\pm y_0 \pm y_1)$	2–3	189	1.00	98.7% / 99.8%
16-input raw-bit gate	$\gamma \sum_{16} s_k x_k$	0–2	19	0.007	0.2% / —
near-dead receiver	mixed $d16$+$d8$, tiny	3	6	0.95	— / 0.04%

to such a gate are raw bits, its Walsh power is spread over many modes, so the fraction visible in the target-tree modes decreases rapidly with width (1/3, 0.15, 0.044, 0.007).

The only genuinely compositional step occurs at the top of the hierarchy. The two degree-8 half-parities are already available, and 8+8 corner detectors combine them to produce χ_{d16}. In each such neuron, the degree-16 product is one of three equal nonconstant Walsh components rather than a dedicated feature. Because the readout follows immediately afterward, the network never needs to distill χ_{d16} into a separate amplifier population.

Therefore, in the meaningful mechanistic sense, "the neurons computing the $d16$ parity" are the 8+8 corner-detector population as a whole. The neurons selected purely by a $d16$-dominance label mostly reflect an argmax over tiny coefficients rather than the true location of the computation.

Artifacts: `gdrive:parity/analysis_output/neuron_mechanisms/wd3_k0_N2048/step_00500000/` (`d16_classes.csv`, figures, this report; companion $d2$/$d4$/$d8$ probes in the same folder).